
Smart Pipeline Failure Prediction System

Technical Interview Q&A; Guide (Hinglish)

1. Core Machine Learning & Data Science

Q1: What is the main objective of this project, and which ML models are used?

Answer: Is project ka main goal Indore city ke water supply network mein underground leaks aur pipe bursts hone se pehle prediction karna hai. Iske liye humne **two Machine Learning models** train kiye hain using **Random Forest**:

1. **Random Forest Regressor:** Yeh model pipe segment ka predicted pressure (PSI) calculate karta hai.
2. **Random Forest Classifier:** Yeh model output probability deta hai ki us pipe mein leakage hone ka risk kitna percent hai (0% to 100%).

Q2: Why did you choose Random Forest instead of Deep Learning or other models?

Answer: Humare pass structured tabular telemetry dataset hai jisme numeric features (flow rate, vibration, temp, coordinates) hain. Tabular data ke liye **Random Forest** algorithm bahut robust hoti hai:

- Isme overfitting risk low hota hai (bagging concept ki wajah se).
- Feature importance nikalna aasan hota hai jisse hum samajh sakein ki leakage ke main factors kya hain.
- Industrial execution ke liye yeh model kafi fast hai aur deep learning ke mukable low hardware specs par run ho sakta hai.

Q3: How did you handle class imbalance in Leakage Prediction?

Answer: Real world pipelines mein, normal conditions jyada hoti hain aur leakage occurrences (< 10%) bahut kam hoti hain, jisse dataset highly imbalanced ho jata hai. Humne ise do tariko se handle kiya:

1. Split karte waqt **Stratified Split** use kiya taaki train aur test split mein leakage data ka ratio same rahe.
2. Model training ke dauran `class_weight='balanced'` parameter set kiya, jisse classifier leakage nodes (minority class) ko detect karne par jyada focus kare.

Q4: Explain the feature engineering you performed for this project.

Answer: Raw telemetry features ke alawa, humne specific hydraulic and physical domain-specific features generate kiye:

- **Wear_Factor:** Operational Hours ko linear wear constant se compute kiya taaki pipeline material fatigue model ho sake.
- **Thermal_Stress:** Temperature, Vibration aur Flow_Rate ko combine karke system tension track karne ke liye.
- **Distance_from_Center:** City center ke main pumping station coordinates se absolute distance measure kiya.
- **Pressure_Flow_Ratio:** Pipe pressure aur flow rate ka ratio anomaly markers detect karne ke liye.

2. Web Development & System Architecture

Q5: How does the real-time simulation system work in the backend?

Answer: Backend Flask app mein humne ek **background daemon thread** generate kiya hai (`run_pipeline_simulation`). Yeh thread har 15 seconds mein background mein sleep se uthata hai aur random pipeline segments ka data update karke real-time leak anomalies simulate karta hai. Jab koi anomaly

high-risk threshold cross karti hai, toh backend system automatically logged-in alert email par secure notification alert bhej deta hai.

Q6: Explain the GIS integration and why you modified the Map engine.

Answer: Map integration ke liye humne **Leaflet.js** framework use kiya hai frontend mein. Pehle default map normal gray vector overlay (CARTO Voyager) dikha raha tha, jo terrain details ke liye relevant nahi tha. Humne customize karke **OpenStreetMap Street view** aur **Esri Satellite terrain view** add kiya aur overlapping markers block hone se bachane ke liye coordinates sort kiye.

Q7: How is SMTP handled, and what changes did you make to the Admin Panel?

Answer: Pehle user ko admin panel UI se host, port, user email, password setup karna padta tha jo secure nahi tha. Humne use completely cleanup kiya. Ab credential configurations (host, port, user) backend mein secure variables ke roop mein store hote hain. Admin user ko ab setting card mein sirf target alert email id dalkar save karna hota hai, aur system automated SMTP secure connection trigger kar deta hai.